

Thomas Yahl

Madison, WI | Thomasyahl@gmail.com | [LinkedIn](#) | [Github](#) | [Personal Webpage](#)

Profile

Analytical and results-driven mathematician with a strong foundation in statistics, data science, and programming. Proven leadership and experience managing teams, developing software, and conducting large-scale data analysis. Seeking to transition into an actuarial role, leveraging quantitative expertise and programming skills to support risk modeling, pricing, and financial forecasting in the insurance industry.

Education

Ph.D. Mathematics, Texas A&M University	(GPA = 3.89)	2023
B.S. Mathematics, Minor Computer Science, Purdue University-Indianapolis	(GPA = 3.68)	2017

Skills and Certifications

- Programming and Software: Python, julia, Excel, SQL, R, Git, scikit-learn, Macaulay2
- SOA Financial Mathematics Exam: Preliminary Pass, October 2025
- SOA Probability Exam: Scheduled, November 2025

Work and Leadership

Van Vleck Visiting Assistant Professor, University of Wisconsin-Madison (UWM) 2023 - Present

- Published 2 articles on machine learning and statistics [[Article1](#), [Article2](#)].
- Organized a mathematics software workshop funded by NSF with over 60 participants.
- Designed and taught courses in calculus, linear algebra, differential equations, and topology.

Graduate Research Assistant, Texas A&M University (TAMU) 2017 - 2023

- Developed open-source mathematics software in julia and Macaulay2 [[Software1](#), [Software2](#), [Software3](#)].
- Published 5 articles in Journal of Algebra and its Applications, Numerical Algorithms, and others.
- Led a team of 3 undergraduate researchers to analyze data for over 40,000 geometric problems.
- Taught qualifying exam prep courses, which aided graduate students in passing qualifying exams.

Assistant Manager, Math Assistance Center 2013 - 2017

- Managed the hiring, scheduling, and evaluation of a team of 30 mathematics tutors.
- Organized exam prep sessions for over 1,500 students, improving pass rates and confidence.
- Provided direct tutoring for undergraduate students in upper-level mathematics courses.

Selected Projects

Macaulay2 Workshop: Madison 2025 – UWM Spring 2025

[\[Webpage, Grant\]](#)

- Awarded \$40,000 NSF grant to host a mathematics software workshop with over 60 participants.
- Expanded the capability of at least 7 mathematics software packages.
- Provided participants with hands-on learning and coding experience.

TAMIDS Data Science Competition – TAMU Spring 2022

[\[Github\]](#)

- Analyzed collaboration trends across departments using data from over 20,000 research articles.
- Used Python, pandas, and scikit-learn for data collection, cleaning, and analysis.
- Awarded 4th place and Best Use of Additional Data.

Decomposable Sparse Systems – TAMU Fall 2020

[\[Slides, Article, Software\]](#)

- Developed an algorithm to improve speed and accuracy in polynomial system solving.
- Achieved nearly 50% performance improvement across over 10,000 test cases in Macaulay2.
- Published results and software in peer-reviewed journals.

Publications, teaching evaluations, and awards are available upon request.